

PROJECT PORTFOLIO - PRINCE PATEL

M.Tech (Transportation Systems Engineering), IISc Bengaluru

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Profile: Mobility research and analytics with emphasis on questionnaire design, experimental design, field surveys, data quality/cleaning, and statistical/econometric modelling (Python; R; Power BI; QGIS; Unity).

1. M.Tech Thesis Project:- Cyclists' Perceived Safety in Indian Urban Context (VR/Simulator Study) [Report Link](#)

Objective: Measure and model how cyclists' perceived safety varies across infrastructure (bicycle lane vs no lane), traffic speed (low/high), and traffic volume (low/high), and relate safety perception to observed riding behaviour.

My Responsibilities (End-to-End):

- **Experimental Design, Pilot Testing & Protocol:** Defined the treatment structure and execution plan; conducted a pilot study to validate the experimental design and protocol; refined instructions/timing to improve repeatability and reduce bias.
- **Questionnaire / Survey Instrument Development:** Designed a structured questionnaire comprising Likert-scale (1–5) perceived safety questions, a brief demographic section, and SSQ items to monitor simulator discomfort and ensure data validity; structured the response capture so ratings are recorded immediately after each condition for better reliability.
- **Data Collection & Validation:** Collected participants' safety ratings and ensured each response was correctly linked to the right traffic condition/treatment; checked the dataset for missing entries and inconsistent responses to keep the data clean and reliable.
- **Data Cleaning & Feature Engineering:** Cleaned simulator trajectory data by handling missing values, removing unrealistic records, smoothing obvious noise, and excluding buffer sections; engineered variables including cyclist speed, lateral/kerb distance, longitudinal position, time-based measures, acceleration/variability indicators, and turning/steering behaviour (where available).
- **Statistical & Econometric Analysis:** Modelled perceived safety ratings using SEM and ordinal econometric models (Ordered Logit / Ordered Probit); tested the significance of key effects (lane, speed, volume) and their interactions; built supporting regression models for behavioural outcomes (speed and lateral/kerb distance) to check behavioural consistency with perceived safety.

Tools: R / Python, statistical modelling, experimental design methods, data wrangling, visualization.

2. CEEW (Sustainable Mobility) - Summer Intern (June–July 2025) — Microbus Feasibility Study (Uttar Pradesh) [Report Link](#)

Objective: Support a pilot assessment of microbus feasibility in small and medium cities by collecting primary data, ensuring data quality, and generating evidence-based insights for service design and implementation.

My Contributions (as part of the project team):

- **Survey Planning & Questionnaire Development:** Drafted and refined structured questionnaires for user perception surveys and operational surveys (IPT drivers and IPT

dealers); ensured clear wording, consistent response options, and analysis-ready question formats.

- **Field Survey Execution & Stakeholder Engagement:** Conducted on-ground surveys with commuters/users, IPT drivers, and IPT dealers; engaged with stakeholders such as traffic police and PWD officials to understand operational constraints and implementation feasibility.
- **Data Cleaning, Validation & Structuring:** Consolidated field survey responses into structured datasets; performed quality checks (missing values, inconsistent entries, duplicate records, coding standardization) to prepare clean, analysis-ready data.
- **Statistical / Econometric Analysis:** Applied statistical and econometric techniques in Python/R (e.g., regression and segmentation style analysis) to identify significant determinants and patterns in user preferences and operational issues; synthesised results into actionable insights relevant to service reliability and route/fleet planning.

Tools: Python/R, survey data processing, statistical analysis, Power BI.

3. Course Project:- Last-mile Data Analysis & Visualization (Bengaluru, Delhi, Nagpur) [Report Link](#)

Objective: Analyse metro last-mile connectivity across three Indian cities and produce interactive dashboards + spatial maps to reveal mode-demographic patterns, travel time, travel cost, travel distance variability, and station-level accessibility gaps for planning use.

Work Done:

- Data cleaning & harmonisation (Python): cleaned station survey datasets by applying quality filters (e.g., zero/invalid last-mile time or distance, same OD pairs, extreme outliers), standardised units, and harmonised categories across cities for consistent comparisons.
- Feature engineering: created analysis-ready variables for last-mile time, cost, waiting time, distance (e.g., TT_{LM}, TC_{LM}, WT_{LM}, TD_{LM}) and derived demographic/mode groupings (age groups, income buckets, and consolidated last-mile mode categories such as NMT/IPT/Private/Bus/Mixed).
- Statistical analysis: ran Chi-square tests to assess association between last-mile mode choice and demographics (age, gender, income), reporting effect size using Cramer's V; applied One-way ANOVA to test whether last-mile travel time differs significantly across modes, including effect sizes (η^2 , ω^2) for interpretability.
- Power BI dashboards: built interactive, multi-page dashboards with slicers (age/income/gender/ownership) and visuals such as mode share distributions, mode vs time comparisons, and station-wise summaries for Bengaluru/Delhi/Nagpur.
- QGIS spatial analysis: joined station-level indicators to maps and identified high access-burden stations using rule-based thresholds (e.g., low walk share and/or high last-mile time), highlighting geographic clusters of last-mile gaps.

Tools: Python, Power BI, QGIS **Methods:** Descriptive stats, Chi-square, One-way ANOVA

4. Course Project:- Bridging First Mile and Last Mile Gaps in Bengaluru's Public Transport Network (Survey Study) [Report Link](#)

Objective: Assess first- and last-mile connectivity barriers for Bengaluru commuters and develop actionable recommendations to improve Metro-BMTC integration and access to transit.

Work Done:

- Survey design & data collection: designed a structured 40-item questionnaire; collected 55 responses.
- Data cleaning & validation: filtered responses to retain regular non-leisure commuters, resulting in 52 valid entries, and prepared a clean analysis-ready dataset.
- Quantitative analysis: produced descriptive summaries and plots for mode shares, first-/last-mile distance ranges, access mode patterns, and mode ratings; highlighted key barriers for first mile, last mile, and BMTC service reliability.
- Cross-tab insights: examined relationships such as main mode vs last-mile method and distance effects.
- Qualitative coding & recommendations: analysed open-ended responses and proposed interventions across infrastructure, feeder enhancements, and bus-metro integration.

Tools/Methods: Google Forms survey, descriptive statistics, cross-tabulations, visualizations, qualitative coding

5. **Course Project:- Traffic Data Collection & Analysis at a Signalized Intersection (Bengaluru)** [Report Link](#)

Objective: Collect and analyse intersection traffic parameters - classified turning volumes, vehicle composition, spot speeds, queue lengths, and signal timings - to assess operational performance and generate structured inputs for modelling and improvement recommendations.

Work Done:

- Field data collection: recorded turning movement counts by vehicle class, spot speeds (30 m trap), end-of-red queue lengths, and phase-wise signal timings.
- Data cleaning & validation: consolidated multiple survey sheets, synchronised timestamps across datasets, and performed consistency checks.
- Performance assessment: summarised demand/composition; computed class-wise speeds; analysed queues and observed cycle/phase timings to interpret residual queues and underserved movements.

Tools/Methods: PTV VISSIM, field survey sheets, video-based turning counts, 30 m speed trap, queue measurement, signal timing analysis

6. **Course Project:- Intermodal Travel Audit (Walk–Bus–Metro) in Bengaluru City** [Report Link](#)

Corridor: IISc (IDR Building) ↔ Mantri Square (Sampige Road) **Modes:** Walking, BMTC bus, Namma Metro

Objective: Evaluate real-world intermodal integration in Bengaluru by documenting a door-to-door trip and assessing transfer experience, information/payment systems, pedestrian safety, and overall travel quality, with actionable recommendations for improvement.

Work Done:

- Field data collection (two-team design): conducted a coordinated corridor audit using two teams taking different mode sequences (bus→metro vs metro→bus).
- Journey time decomposition: recorded timestamps for access, wait, in-vehicle travel, transfers, and ticketing steps; compared observed vs app-estimated travel times.
- Payment & information assessment: evaluated ticketing/payment usability and wayfinding and real-time information.
- Safety & walk-quality audit: documented pedestrian infrastructure gaps and risks along access/egress paths.
- Comparative insights & recommendations: compared cost–time trade-offs and proposed practical improvements (bus stop design, crossings, lighting, sidewalk continuity, station-area management).

Tools/Methods: Time-stamp logs, structured field-note sheets, photos/videos, map-based route tracking, cost/time decomposition (IVTT/OVTT)

7. **Course Project:- Travel Mode Choice Modelling (Cluj-Napoca, Romania) using Discrete Choice Models** [Report Link](#)

Objective: Model home-to-work commuter mode choice in the Cluj-Napoca Metropolitan Area and quantify how level-of-service (travel time, travel distance) and socio-demographic factors influence the probability of choosing Walk / Public Transport / Private Transport.

Work Done:

- Problem framing & literature: structured determinants of mode choice (Level of Service + socio-demographics + psychological/attitudinal factors).
- Data processing: prepared Revealed Preference and Stated Preference (RP/SP) style dataset; cleaned and recoded variables; built mode-specific attributes (travel time, distance, etc.) and traveller characteristics.
- Model development (MNL): specified utility functions and estimated parameters via maximum likelihood; interpreted behavioural implications.
- Advanced model (MMNL): extended to Mixed Logit to capture preference heterogeneity and compared against MNL.
- Policy simulations: evaluated scenario impacts on predicted mode shares (e.g., PT travel time reduction and targeted fare subsidy for low-income users).

Tools/Methods: R (MLE estimation), Multinomial Logit (MNL), Mixed Multinomial Logit (MMNL), policy scenario forecasting

8. **Course Project:- Smart Scientific Contributor Verification, Supervised Machine Learning** [Report Link](#)

Objective: Build a metadata-driven authorship verification framework to classify whether a candidate author is a genuine contributor or a non-contributor for a research article.

Work Done:

- Built a supervised machine learning pipeline for binary classification of contributor vs non-contributor using article metadata.
- Performed data preprocessing, feature engineering, and preparation of structured training datasets.
- Evaluated Logistic Regression, Support Vector Machine (SVM), Random Forest, and XGBoost models.
- Applied hyperparameter tuning and 10-fold stratified cross-validation, including regularization-based refinement.
- Compared classification performance across models and found XGBoost to be the best-performing model.

Tools/Methods: Python, Scikit-learn, XGBoost, supervised learning, feature engineering, stratified cross-validation

9. **Course Project:- Industrial Predictive Maintenance using Binary & Multi-Label Machine Learning** [Report Link](#)

Objective: Develop a predictive maintenance framework for industrial equipment to identify machine failure and diagnose multiple fault categories using sensor telemetry data.

Work Done:

- Developed a feature-engineered sensor-telemetry pipeline for binary failure detection and multi-label fault diagnosis.
- Processed industrial machine data using scaling, encoding, and structured preprocessing steps for model development.
- Built models for failure and fault-category prediction across TWF, HDF, PWF, OSF, and RNF classes.
- Evaluated Logistic Regression, SVM, Random Forest, ANN, LightGBM, and XGBoost models.
- Applied SMOTE-based stratified cross-validation and threshold tuning to improve performance under class imbalance.
- Compared model outputs and found XGBoost to be the best-performing model.

Tools/Methods: Python, Scikit-learn, XGBoost, LightGBM, ANN, SMOTE, predictive maintenance, binary classification, multi-label learning

10. **Course Project:- Power Grid Cyber-Security using Unsupervised Machine Learning** [Report Link](#)

Objective: Design an unsupervised intrusion detection framework for smart-grid communication systems using IEC 61850 GOOSE traffic data.

Work Done:

- Engineered temporal and protocol-consistency features from IEC 61850 GOOSE traffic, including inter-arrival deviation, heartbeat variation, sequence/state transitions, and rule-violation indicators.

- Built an unsupervised anomaly-detection pipeline for identifying suspicious activity in power-grid communication traffic.
- Evaluated Local Outlier Factor (LOF), Isolation Forest, One-Class SVM, Gaussian Mixture Models (GMM), K-Means, and DBSCAN.
- Assessed models for both binary intrusion detection and multi-class attack characterization.
- Compared performance across methods, where LOF performed best for binary detection and K-Means performed best for multi-class characterization.

Tools/Methods: Python, unsupervised learning, anomaly detection, clustering, LOF, Isolation Forest, One-Class SVM, GMM, K-Means, DBSCAN

11. **Course Project:- Atliqo Bank Credit Card Issuance and Customer Segmentation** [Report Link](#)

Objective: Analyze customer transaction behavior and credit usage patterns to identify under-served customer groups and support targeted credit card issuance strategy.

Work Done:

- Analysed 50,000 transaction records across 1,000 customers to study spending behaviour and credit usage patterns.
- Performed data cleaning, preprocessing, and exploratory analysis to derive customer-level behavioural insights.
- Identified credit card underutilization among the 18–25 age group through segmentation-based analysis.
- Designed a targeted credit card issuance strategy for the identified customer segment.
- Evaluated the proposed strategy using a 2-month A/B experimental framework.

Tools/Methods: Python, exploratory data analysis, customer analytics, segmentation, A/B testing, data visualization